

Maxwell Equation (在 vacuum, source free)

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{cases}$$

對 Faraday's law 取 curl

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) \quad - ①$$

由向量恆等式

$$\nabla \times (\nabla \times \vec{A}) = \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} \quad - ②$$

把 ① 代入 ② 可得

$$\nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} (\nabla \times \vec{B})$$

$$\Rightarrow \nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = 0 \quad - ③$$

因為想找的是沿著 z 方向傳播的 time-harmonic field, 所以試

$$\vec{E}(x, y, z, t) = \hat{e}_i E_0 U(x, y) e^{-j(\omega t - \beta z)} \quad - ④$$

$\hat{e}_i$: 極化方向

E_0 : 最大振幅

$U(x, y)$: field

把 ④ 代入 ③

$$\hat{e}_i A = \hat{e}_i E_0 e^{-j(\omega t - \beta z)}$$

$$A \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) U(x, y) + \frac{\partial^2}{\partial z^2} E - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = 0$$

$$\nabla^2 (\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}) U(x, y) - K^2 U(x, y) + \mu_0 \epsilon_0 \omega^2 U(x, y) = 0$$

$$\nabla^2 U(x, y) + (\mu_0 \epsilon_0 \omega^2 - \beta^2) U(x, y) = 0$$

在真空中的波數 $k_0 = \frac{\omega}{c}$. 而 $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$\text{所以 } k_0^2 = \mu_0 \epsilon_0 \omega^2 = \frac{2\pi}{\lambda_0}$$

對於有 $n(x, y)$ 的折射率分佈下假設偏振方向固定、且用單-scalar field $U(x, y)$ 來描述 mode, 會變成 $(\mu_r \epsilon_r)(\mu_0 \epsilon_0 \omega^2) = n^2 k_0^2$

$$\Rightarrow \nabla^2 U(x, y) + (k_0^2 n^2 - \beta^2) U(x, y) = 0 \quad - (5)$$

這個稱為 Helmholtz 方程 (波動方程在頻率固定下的版本)
是接下來整個 Finite Difference solver 的核心公式喔!

什麼是 β (propagation constant)?

它是光波沿傳播方向 (z) 的空間振盪頻率, 也就是 z 方向的波數 k_z , β 的大小決定光前進單位距離後, 相位累積的多寡

什麼是 n_{eff} (effective index)

在均勻的介質下, $k = n k_0$, 所以在傳播方向上也應有 $\beta = n_{\text{eff}} k_0$.
代表在 z 方向上的折射率, 所以 $n_{\text{eff}} = \frac{\beta}{k_0}$

要如何把剛才的式 ⑤ 化成電腦可以用的形式呢?

其實這個方程是一個特徵值問題喔!

$$\nabla^2 U(x,y) + (n^2 k_0^2 - \beta^2) U(x,y) = 0$$

$$\left(\frac{\nabla^2}{k_0^2} + n^2\right) U(x,y) = \frac{\beta^2}{k_0^2} U(x,y) = n_{\text{eff}}^2 U(x,y)$$

$$\Rightarrow \left(\frac{\nabla^2}{k_0^2} - n^2\right) \underbrace{U(x,y)}_{\substack{\downarrow \\ \text{eigenvector}}} = \overset{\text{eigenvalue}}{n_{\text{eff}}^2} \underbrace{U(x,y)}_{\substack{\downarrow \\ \text{eigenvector}}} \quad - (6)$$

那如果可以化成某種矩陣, 就可以讓電腦很好的算!!

Finite Difference Method (有限差分法)

把空間切成格點, 用格點的差商作為函數的微分近似

把 PDE 化為代數方程。

所以我可以去找 $U(x+x, y)$. $U(x-x, y)$. $U(x, y+y)$. $U(x, y-y)$

整理起來, 就能用相鄰的格點計算

利用 Taylor Expansion: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$

在 x 上展開, 找 $f(x+\Delta x)$

$$\Rightarrow f(x+\Delta x) = f(x) + f'(x)\Delta x + \frac{f''(x)}{2!}(\Delta x)^2 + \dots$$

對 $U(x, y)$

$$\begin{cases} U(x+\Delta x, y) = U(x, y) + \Delta x \frac{\partial U(x, y)}{\partial x} + \frac{(\Delta x)^2}{2} \frac{\partial^2 U(x, y)}{\partial x^2} + \dots \\ +) \quad U(x-\Delta x, y) = U(x, y) - \Delta x \frac{\partial U(x, y)}{\partial x} + \frac{(\Delta x)^2}{2} \frac{\partial^2 U(x, y)}{\partial x^2} - \dots \end{cases}$$

$$\frac{\partial^2 U(x, y)}{\partial x^2} = \frac{U(x+\Delta x, y) - 2U(x, y) + U(x-\Delta x, y)}{(\Delta x)^2}$$

$y+\Delta y$ 同理

$$\frac{\partial^2 U(x, y)}{\partial y^2} = \frac{U(x, y+\Delta y) - 2U(x, y) + U(x, y-\Delta y)}{(\Delta y)^2}$$

Simplify notation: $\begin{cases} U(x, y) = U_j^i \\ U(x \pm \Delta x, y) = U_{j \pm 1}^i \\ U(x, y \pm \Delta y) = U_j^{i \pm 1} \end{cases} \Rightarrow \begin{cases} \frac{\partial U(x, y)}{\partial x^2} = \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{(\Delta x)^2} \\ \frac{\partial U(x, y)}{\partial y^2} = \frac{U_j^{i+1} - 2U_j^i + U_j^{i-1}}{(\Delta y)^2} \end{cases}$

代回前面的式 (6), 並令 $\Delta X^2 = F_0^2 (\Delta x)^2$, $\Delta Y^2 = K_0^2 (\Delta y)^2$

$$\frac{U_j^{i+1} - 2U_j^i + U_j^{i-1}}{(\Delta x)^2} + \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{(\Delta y)^2} + (N_j^i)^2 U_j^i = N_{\text{eff}}^2 U_j^i$$

整理一下：

$$\frac{U_j^{\hat{i}-1}}{(\Delta X)^2} + \frac{U_{j-1}^{\hat{i}}}{(\Delta Y)^2} - \left(\frac{1}{(\Delta X)^2} + \frac{1}{(\Delta Y)^2} - \tilde{N}_j^{\hat{i}} \right) U_j^{\hat{i}} + \frac{U_j^{\hat{i}+1}}{(\Delta X)^2} + \frac{U_{j+1}^{\hat{i}}}{(\Delta Y)^2} = \tilde{N}_{\text{eff}}^{\hat{i}} U_j^{\hat{i}} \quad (6)$$

這裡有了上下左右的5種格子，想辦法寫成Matrix就大功告成了

$$\text{令 } \begin{cases} a_j^{\hat{i}} = \tilde{N}_j^{\hat{i}} - \frac{1}{(\Delta X)^2} - \frac{1}{(\Delta Y)^2} \\ b = \frac{1}{(\Delta Y)^2} \end{cases} \quad \begin{cases} (\Delta X)^2 = k_0^2 (\Delta x)^2 \\ (\Delta Y)^2 = k_0^2 (\Delta y)^2 \end{cases}$$

$$\underbrace{\frac{U_j^{\hat{i}-1}}{(\Delta X)^2}}_{\text{都是 } \hat{i}-1} + \underbrace{\left(b U_{j-1}^{\hat{i}} + a_j^{\hat{i}} U_j^{\hat{i}} + b U_{j+1}^{\hat{i}} \right)}_{\text{都是 } \hat{i}} + \underbrace{\frac{U_j^{\hat{i}+1}}{(\Delta X)^2}}_{\text{都是 } \hat{i}+1} = \tilde{N}_{\text{eff}}^{\hat{i}} U_j^{\hat{i}}$$

所以如果把 y 方向都包起來： $U^{\hat{i}} = \begin{bmatrix} U_1^{\hat{i}} \\ U_2^{\hat{i}} \\ \vdots \\ U_J^{\hat{i}} \end{bmatrix}$

就可以寫成 matrix

$$B U^{\hat{i}-1} + A U^{\hat{i}} + B U^{\hat{i}+1} = \tilde{N}_{\text{eff}}^{\hat{i}} U^{\hat{i}}$$

其中：

$$A = \begin{bmatrix} a_1^{\hat{i}} & b & 0 & \cdots & 0 \\ b & a_2^{\hat{i}} & b & \cdots & 0 \\ 0 & b & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & a_J^{\hat{i}} \end{bmatrix} \rightarrow \begin{array}{l} \text{主對角線是 } a_j^{\hat{i}} \\ \text{兩副對角線是 } b \end{array}$$

$$B = \frac{1}{(\Delta X)^2} I$$

↳ identity matrix

為什麼 A 的 row 1 不是 $b, a, b, \dots$ 呢?

因為我們的 boundary condition 設定成表面為 0

$$\text{因此 } U_0^i, U_{i+1}^i = 0$$

最後, 把 x 方向也包起來...

$$M U = N_{\text{eff}} U$$

其中

$$M = \begin{bmatrix} A^1 & B & 0 & \dots & 0 \\ B & A^2 & B & \dots & 0 \\ 0 & B & A^3 & \dots & \\ \vdots & \vdots & \dots & \ddots & \\ 0 & 0 & \dots & \dots & A^N \end{bmatrix} \begin{array}{l} \rightarrow \text{main diagonal } A^i \\ 2 \text{ secondary diagonal } B \end{array}$$

$$U = \begin{bmatrix} U^1 \\ U^2 \\ \vdots \\ U^N \end{bmatrix}$$

這樣就可以把折射率分佈的矩陣放入, 建構好後

就有了 eigenvalue : N_{eff}^2

eigenvector : $U(x, y)$